Literature position for the threshold-entry contest

Survey and verified bibliography

1 September 2026, revised 14 October 2026

Note on references to `PROOFS.tex`. The proofs document `PROOFS.tex` was retired on 7 October 2026 (pass 6 of `TODO.md`). Its results are the model rows M1 to M27 of `MANUSCRIPT.tex`, its non-proof content is in that document's appendices, and its section on the contribution is §18 below. Where this survey says what `PROOFS.tex` states, it records that document as it stood, which can be recovered from git at commit `7132717f`. `RETIREMENT.md` lists where each part of it went.

1 Purpose and verification standard

This document settles the literature check flagged as the gating open item in `PROOFS.tex` (§Open items, item 4) and in the referee handover (R1 Comment 4). The question it answers is narrow: is the mechanism claimed as novel — an identical nominal entry cost c made asymmetric in utility by concavity, so that *who* enters a fixed-prize contest is determined by the wealth distribution rather than by ability — already occupied?

Every entry in `refs.bib` has author list, title, journal, year, volume and pages verified against at least one of: the article's own title page; the publisher's article page; or the reference list of a published peer-reviewed article. Where a claim below describes what a paper *does* rather than what it is, the evidence level is marked:

[F] full text read;

[A] abstract or publisher summary only — model details are reported as the abstract states them and have not been checked against the body;

[C] characterised via a citing paper's description, not the source itself.

Where the text read was a working paper or draft rather than the published article, the tag says so — "[F, WP version]" or "[F, draft version]" — and proposition numbers quoted from it may not match the published version.

Claims marked [A] or [C] are not safe to cite as statements about what a paper proves. They are safe as statements about what a paper is about. Anything that enters `PROOFS.tex` as a positioning claim should be upgraded to [F] first.

2 The closest paper on the affordability channel, read in full

Schroyen and Treich (2016) is the nearest occupant of the mechanism, and the one against which the contribution has to be stated. It has now been read in full **[F]**. The version read is TSE Working Paper 16-699, whose cover is dated September 2016 and which carries the post-review *Games and Economic Behavior* manuscript dated 11 April 2016. The published article replaced

it; the published text has not been checked line by line against it, so any quotation of a theorem number should be confirmed there before submission.

Correction to the previous draft of this document. An earlier version of this survey, written from the abstract, stated that their setting produces no comparative statics on the wealth distribution. That is wrong. Their Theorem 3 signs the effect of a mean-preserving spread in wealth in the privilege contest. The overlap with P9 is therefore real and is set out below.

Primitives. A static, one-shot, two-player game. Appendix A.1 states that the only source of heterogeneity is wealth w_i. Three models are distinguished by whether the rent r and the effort x_i are commensurable with wealth inside u:

$$\begin{aligned} \text{privilege:} \quad & U_i = u(w_i - x_i) + \pi_i r, \\ \text{ability:} \quad & U_i = \pi_i u(w_i + r) + (1 - \pi_i)u(w_i) - c(x_i), \\ \text{rent-seeking:} \quad & U_i = \pi_i u(w_i + r - x_i) + (1 - \pi_i)u(w_i - x_i), \end{aligned}$$

with π the power-logistic Tullock CSF of decisiveness m. Effort is continuous on a convex feasible set. Dynamic effects are listed in their conclusion as an extension not undertaken, as is an arbitrary number of players.

Consequence 1: the wealth/income distinction does no work. Because the model is static with no saving, w_i is a one-period resource level out of which effort is paid. That is formally the same object as the resource variable in `PROOFS.tex`. Describing our primitive as income and theirs as wealth is a labelling difference between two papers, not a difference between two models, and it cannot be used to separate them. The separation has to come from structure.

Consequence 2: our model is a privilege contest, confirmed from the functional form. Entry cost c is paid out of resources and V is separable and exogenous, so only the marginal-cost channel is live and the marginal-benefit channel is switched off by construction. This is now verified against $U_i = u(w_i - x_i) + \pi_i r$ rather than inferred from the taxonomy's name. It also fixes the open design question from the rebuild handover: making V money added to the resource would move the model to their rent-seeking contest, where the two wealth effects oppose and, under CARA or quadratic utility, exactly cancel (pp. 15 and 18).

Consequence 3: the P9 overlap, stated honestly. Their Theorem 3 gives, for the symmetric privilege contest, that a small mean-preserving spread in wealth raises aggregate effort if and only if

$$2A(1 - m^2) > P, \qquad A = -\frac{u''}{u'}, \qquad P = -\frac{u'''}{u''},$$

reducing to $m < 2^{-1/2} \approx 0.707$ under CARA, to $m < 1$ under quadratic utility, and to $\gamma(\frac{1}{2} - m^2) > \frac{1}{2}$ under CRRA with coefficient γ. So a mean-preserving spread in the resource distribution, inside a privilege contest, is already in print. The displays here use our notation, π_i and γ, where the source writes Π_i and ρ.

The pair used in `PROOFS.tex` to show the dependence on u''', CARA with $a = 1$ against log utility at $w = 1$, shares $A = 1$ and differs in P (1 against 2). The condition therefore separates the pair exactly when $1 < 2(1 - m^2) < 2$, that is for $m \in (0, 2^{-1/2})$, and at $m = 4/5$ both members give the same sign (`lit/schroyen_treich_2016`, ST-12). The contrast should be stated with that range, or with a pair chosen for the m at hand.

Consequence 4: what survives, and why it is the contribution. Their result is local and heavily conditioned. It is a second-order effect of a small spread evaluated at a *symmetric* two-player equilibrium, obtained as a quadratic form in the Hessian of the best-response function (their Theorem 2), and its sign turns on the CSF decisiveness parameter and on the third derivative of u. Their own summary concedes that the wealth-inequality effects are the most complex in the paper and that signing a mean-preserving spread is very difficult without further restricting functional forms. The condition $u \in \Omega$ they need in the rent-seeking case — that a mean-preserving spread of background risk raises absolute risk aversion — is, by Eeckhoudt et al. (1996), a restriction on higher-order risk attitudes, and is stronger than DARA, being related to the risk vulnerability of Gollier and Pratt (1996).

P9-gen requires none of this. Its sign rule needs only that κ is strictly decreasing and that T is a pivot-spread, with the flip located at $w_{(k^*)}$ against the pivot x_0. No third derivative, no prudence condition, no decisiveness parameter, no functional-form restriction, and the statement is global rather than a local second-order expansion. The contribution sentence should therefore be:

> Moving the choice from the intensive to the extensive margin makes the mean-preserving-spread comparative static sign-determinate without higher-order restrictions on utility.

That is a sharp claim stated directly against a named published result, and it is stronger than anything the mechanism alone supports.

Consequence 5: a caveat that narrows the noise argument. They note, citing Jia (2008), that the power-logistic Tullock CSF can be derived as a rank-order tournament in which the noise terms are inverse-exponential, and remark that their contests may therefore also be read as rank-order tournaments. The deterministic-versus-noisy contrast drawn against Hopkins and Kornienko (2010) in §3 therefore does *not* extend to Schroyen–Treich. The remaining difference is narrower but real: their noise is one specific family, whereas F and G here are arbitrary subject only to the stated regularity.

Consequence 6: no entry margin. There is no participation decision anywhere in the paper. The zero-effort discontinuity in the best-response functions is a corner of the continuous effort choice, not a fixed-cost entry decision. The extensive margin is genuinely absent, which is what P5 and P7 require and what makes the contribution sentence above available.

A connection worth checking. They contrast their mean-preserving-spread results with the neutrality result of Bergstrom et al. (1986) for private provision of public goods, and note in a footnote to their Theorem 2 that the quadratic form vanishes under the Bergstrom–Varian conditions under which the distribution of agent characteristics has no effect on aggregate effort. Our anonymity proposition — that Δ depends on the entrant count alone and is invariant to the resource distribution — is a neutrality statement of the same shape, and it is precisely what makes P9 tractable. Whether the two are formally related, or merely rhyme, is not established here and should not be asserted without proof.

3 Hopkins and Kornienko: the standard result, read in full

Hopkins and Kornienko (2010) is the paper Referee 2 invoked when predicting that the standard result — inequality reduces status expenditure — would survive in this setting. It has now been read in full **[F]**, and the reading changes the positioning in four specific ways.

Primitives. A continuum of contestants with endowment z drawn from G, private information. Each divides z between visible performance x and private consumption $y = z - x$. A continuum of indivisible rewards with values distributed H is assigned assortatively by rank in performance, so that in a separating equilibrium $G(z) = F(x) = H(s) = r$. Utility is $U(x, y, s)$ with monotonicity, complementarity and own-concavity conditions. The unique symmetric separating equilibrium solves a differential equation with the non-competitive optimum as boundary condition, and equilibrium performance strictly exceeds the cooperative level everywhere above rank zero, so the equilibrium is Pareto dominated from the contestants' point of view. Existence and uniqueness are obtained by fitting the model into Mailath (1987).

Point 1: there is no extensive margin. Every contestant competes. The decision is how much performance to buy, never whether to compete at all. No participation constraint, no entry cost, no fixed fee. The binary entry decision that generates k^* in P5 has no counterpart in the paper. This is the cleanest available statement of what the rebuilt model adds and it is now verified rather than inferred.

Point 2: the contest is perfectly discriminating. The authors note explicitly that much of the surrounding contest literature assumes a partly stochastic award mechanism, and describe their own model by contrast as a perfectly discriminating rank-order tournament. The rebuilt model is on the stochastic side: scores are $\mu r_i + (1 - \mu)s_i e_i$ and the induced CDFs F and G carry all the content. The two models sit in different branches, and P-MU — where the sign of $\partial\Delta(0)/\partial Q$ turns on a hump-shaped weighting — is an object that cannot arise without noise.

Point 3: the wealth channel is a budget, not a fee. Endowment enters as the resource divided between performance and consumption. It sets how much performance a contestant can afford at the margin. It does not act through the utility cost of a fixed payment, which is the entire content of $\kappa(w) = u(w) - u(w - c)$ in the rebuild. These are different mechanisms and should not be described as the same one.

Point 4 — the most useful finding: the right stochastic order is already named. The comparative statics are conducted in the *dispersive order* and the *star order*, not in mean-preserving spreads. Their Definition 1, taken from Shaked and Shanthikumar (2007), is that F is less dispersed than G, written $F \leq_d G$, whenever $G^{-1}(r) - F^{-1}(r)$ is weakly increasing on $(0, 1)$; equivalently, at a fixed rank the more dispersed distribution is the less dense one. They also record that there is no clear relation between the dispersive order and second-order stochastic dominance, because SOSD compares distributions on both mean and dispersion while the dispersive order concerns dispersion alone.

This bears on open item 2 of `PROOFS.tex`, which observes that Theorem `P9gen` requires T to be a pivot-spread and that a general Rothschild–Stiglitz spread need not be monotone, leaving the general case open. The dispersive order is exactly the tool that literature uses for this difficulty, and the two conditions are close. Writing $T = \Lambda_\beta^{-1} \circ \Lambda_\alpha$, the dispersive order says $T(w) - w$ is increasing; if in addition $T(w) - w$ changes sign, it does so once, at a point x_0, giving $T(w) \geq w$ above x_0 and $T(w) \leq w$ below it — which is the pivot-spread hypothesis. So the pivot-spread class is not an ad hoc restriction invented to make the proof work; it is, up to the crossing condition, the standard dispersion order of this literature. That is a considerably stronger position than the current text claims for itself, and it should be checked formally and then stated.

Direction of their result, stated carefully. Under rank indexing, their Proposition 4 assumes the minimum endowment is higher ex post, endowments are less dispersed ex post,

and the maximum endowment is lower ex post. The conclusion is that performance is *higher* ex post on $[0, \hat{r}]$, where $\hat{r}$ is the single crossing point of the two endowment functions, and that utility rises at the bottom but falls for the middle and top. So less dispersion raises competition at the bottom and middle. Read in the other direction, more dispersion of endowments lowers performance there — which is the "standard result" Referee 2 had in mind, and it does hold.

Two cautions follow for P9. First, their conclusion is stated on an interval below a crossing point, not globally, so it is not a uniform-sign aggregate claim and should not be described as one. Second, their model already contains a crossing point at which effects change sign, though it is a crossing in the *endowment functions* with a corresponding sign change in *utility*, not a sign change in the direction of the expenditure response. P9's pivot rule is a different object: a sign reversal in $\mathrm{d}k^*/\mathrm{d}(\text{spread})$ located at the pivot x_0 relative to the marginal entrant's wealth $w_{(k^*)}$. The claim in `PROOFS.tex` §11 that the result "reconciles" the standard result with the motivating pattern is defensible only if stated at that level of precision. As drafted it is too loose, because it invites the reading that the standard result is uniform in sign, which the source does not say. §10 records a second limit, found in the 1992 source of the standard result itself, which shares the direction of the effect with the entry model but not its mechanism.

Genealogy of the dispersive order, corrected. Hopkins and Kornienko (2004) and Hopkins and Kornienko (2009) have now been read in full **[F]**, and together they change what was said about the dispersive order in an earlier draft of this section. The claim there was that the order traces to the general stochastic-orders textbook of Shaked and Shanthikumar (2007) and merely happens to fit this problem. That understates it. The dispersive order, applied to exactly this class of problem, is Hopkins and Kornienko's own methodological innovation, developed across these two papers before the 2010 paper uses it.

Hopkins and Kornienko (2004) is the earlier, income-indexing treatment: agents are indexed by income z itself, and equilibrium strategies and utilities are written as functions of z. Comparative statics there use first- and second-order stochastic dominance, together with two likelihood-ratio refinements — the monotone likelihood ratio (MLR) order and, for spreads, the unimodal likelihood ratio (ULR) order. Under the ULR order the paper's Proposition 4 partitions the population into three zones by two threshold points $\hat{z}_-, \hat{z}_+$ — poor, middle, and rich — and signs the effect of a spread separately in each zone, with the middle zone (where the density ratio is highest) unambiguously seeing higher conspicuous consumption. This is a *two*-crossing-point result, more general than a single pivot.

Hopkins and Kornienko (2009) is explicit, in its own introduction, about why a second methodology was needed: the income-indexing approach of the 2004 paper cannot compare distributions with different supports, so it cannot handle a policy that raises the floor of the income distribution while taxing the top. The paper switches to indexing agents by *rank* $r = G(z)$ instead, and introduces the dispersive order for this purpose, citing Shaked (1982) directly (full citation added to `refs.bib` below). The order is older than that citation. Lewis and Thompson (1981) define it as "ordering in dispersion" by quantile spacing (p. 77), read in full **[F]** (`lit/lewis_thompson_1981`), and their definition is the same order as Hopkins and Kornienko's Definition 1 for continuous, strictly increasing distribution functions (`LewisThompson.cdf_iff_spacing`, `LewisThompson.spacing_iff_diff`, proved in Lean). Their "dispersive" distributions, the log-concave ones of their Theorem 8, are a different notion from a dispersive spread. Rank-indexing plus the dispersive order is presented there as a new technique built specifically to sign comparative statics of a competitive equilibrium under a change in the underlying distribution, which is precisely the class of problem P9-gen belongs to.

Structurally, Hopkins and Kornienko (2009) Propositions 4 and 5 are close cousins of P9-gen.

Proposition 4 gives a sufficient condition for a full Pareto improvement: the poorest agent no worse off, together with the distribution becoming *more* dispersed in their sense, is enough for utility to rise almost everywhere. Proposition 5 is the mirror image: the poorest richer, the richest poorer, and the distribution *less* dispersed, then utility falls for the middle and upper ranks — this is the source proposition the 2010 paper's Proposition 4 (discussed above in §3) generalises. In both cases the sign of the welfare effect is pinned by a single crossing point $\hat{r}$ of the two distribution functions, in exactly the way P9-gen pins the sign of the entry-count effect by the pivot x_0. The two constructions are not the same theorem — theirs crosses in the distribution functions and signs a welfare comparison, P9-gen crosses in $T(w) - w$ and signs an entry-count comparison — but they are the same technique applied to structurally analogous problems, and P9-gen should cite this lineage rather than only the general textbook.

A concrete open question this raises. Hopkins and Kornienko (2004)'s ULR result permits *two* crossing points and three sign regions, not one. P9-gen's pivot-spread hypothesis is a single-crossing object. Whether a two-pivot generalisation of P9-gen exists — for spreads that compress agents in the middle of the wealth distribution while still expanding the tails, which a single pivot cannot represent — is an open question raised directly by this reading and is not addressed anywhere in the current bundle.

Consequence for the income/wealth question. Both papers work entirely in *income* space: the primitive is z, described throughout as income, divided in a single period between a positional good and a non-positional good, with no saving and no mention of wealth as a stock. This is independent confirmation, from the second major reference paper in this literature (after Schroyen–Treich), of the point made when the primitive in `PROOFS.tex` was discussed: a one-period resource divided between an entry cost and residual consumption is standard practice as "income" in this literature, not "wealth," and the field's own vocabulary supports the relabelling already adopted.

No entry margin, confirmed again. As with Hopkins–Kornienko (2010) and Schroyen–Treich, there is no participation decision in either paper: $x \in [0, z/p]$ is a continuous choice on a convex feasible set. The extensive-margin gap the entry_contest bundle fills is confirmed a third time.

Hopkins (2012) and Hopkins (2008) are adjacent, not yet read in full. **[C]**

4 Endogenous entry into contests, read in full

Four papers in this strand have now been read in full **[F]**: Levin and Smith (1994) and Morgan et al. (2012) in their published versions, Fu et al. (2015) in its published version, and Fu and Lu (2010) as MPRA Paper 945 (November 2006), the working version of the *Economic Inquiry* article. Together they settle the entry-structure question more sharply than the abstracts allowed.

The identity problem, stated by the literature itself. All four papers assume identical potential entrants. Every one of them then confronts the same difficulty: with identical agents, the pure-strategy count is determined but the identity of who enters is not. Levin and Smith (1994), p. 586 (main text; footnote 2 is attached to the preceding sentence and concerns Engelbrecht-Wiggans (1987), Hausch (1988) and Harstad (1990)), say of the prior literature's deterministic asymmetric equilibria that the process by which potential bidders divide into entrants and non-entrants is not explained. Morgan et al. (2012), p. 441, say in their own model that because all players are identical the identity of those opting in is not uniquely determined.

The literature has three devices for this. Levin and Smith restore symmetry by mixed entry, so the number of entrants becomes stochastic (binomial). Fu and Lu (2010) (footnote 6) and Morgan et al. (2012) (Proposition 1) use sequential entry with observability, so arrival order breaks the tie; Morgan et al. add private random delays to make the order well defined. Fu et al. (2015) take the mixed route and, because entrants do not observe the number of rivals, need Dasgupta–Maskin's theorem for two-dimensional discontinuous games to establish existence (their Theorem 1).

P5 is the fourth device, and it differs from the other three in kind: the resource ordering $\kappa(w_{(1)}) < \kappa(w_{(2)}) < \cdots$ is a property of the *agents*, not of a timing protocol or a randomisation, so the pure-strategy count k^* is deterministic under *simultaneous* entry with no mixing and no arrival order — and, by the count-invariance proposition in `PROOFS.tex`, the same in every pure-strategy equilibrium.

Morgan et al. (2012) (p. 442) also cite Fullerton and McAfee (1999), who "study contests where agents are heterogeneous and entry is determined by an auction". That paper has now been read in full **[F]**, printed pages 573 to 603 of the published article (`lit/fullerton_mcafee_1999`). Fullerton and McAfee's contestants differ in research cost or in a general quality attribute. The efficient entry equilibrium admits the lowest-cost firms (their Theorem 2, p. 580), and their contestant selection auction admits the best types (Theorem 5, p. 591). Entrant identity is therefore already a property of the agents in that paper, and "differs in kind" narrows to a comparison with the three devices above. What still separates P5 from Fullerton and McAfee is the sorting variable. A Fullerton–McAfee type enters the contest itself, so an entrant's profit depends on its rivals' types (their eq. (4), p. 579). In our model wealth enters only the utility cost of the fee, and the gain depends on the number of entrants alone. That difference carries count invariance. In the Fullerton–McAfee entry stage the number of entrants differs across equilibria. With costs $(2.1, 2.3, 2.5, 2.6)$, prize 1 and fixed cost $19/250$, a set of two firms and a set of three firms are both entry equilibria (`FullertonMcAfee.two_sizes`, proved in Lean). Their Lemma 1 (p. 579) is also a precedent for entry equilibria that exclude a cheaper firm, with the excluded firm's cost at least $(m^2 - m)/(m^2 - m + 1)$ times an entrant's.

The claim should stop there, and an earlier version of this survey did not. Heterogeneous costs do not by themselves pin *which* agents enter: with $\Delta = (12, 10, 1)$ and $\kappa = (2, 5, 8)$ the sets $\{1, 2\}$, $\{1, 3\}$ and $\{2, 3\}$ are all equilibria, one of them with the richest challenger absent. What is delivered is the count, plus existence of the assortative equilibrium, plus its uniqueness under a stated extra condition; absent that condition, assortativity is a selection. That is still a real advance on mixing, arrival order and random delays, each of which leaves even the assortative outcome undefined — but it is a smaller claim than "identity is a property of the agents", and the two footnotes cited above are the places to hang the smaller one.

Levin–Smith. N identical risk-neutral potential bidders, fixed entry cost c paid ex ante before learning one's value (contrast Samuelson's interim costs), n revealed before bidding. Their n^* is defined as the largest integer at which the marginal entrant's expected gain is non-negative — the same object as k^* for identical agents — and their symmetric equilibrium is in mixed strategies with $n \sim \text{Binomial}(N, q^*)$. Two results bear on the open welfare item (R2 Comment 3). In common-value auctions free entry is *excessive* (Proposition 3). The proof runs through the optimal entry fee $e^* > 0$, and the paper then interprets the result as business stealing, citing Mankiw and Whinston (1986) (p. 590). The seller taxes entry; in independent-private-value auctions free entry is *optimal* because the marginal entrant's private gain equals the social gain, via $V_n - W_n = n(V_n - V_{n-1})$ (Proposition 6, equation 18).

Which branch applies is settled by a *criterion*, not by resemblance. Equation (18) is sufficient for Proposition 6, and the paper states no converse. Equation (18) requires the marginal entrant's

private gain to equal the increment to total value. With a prize fixed at V we have $V_n \equiv V$, so $V_n - V_{n-1} = 0$: the social gain from any entrant beyond the first is *exactly zero* while the social cost is $c > 0$. The paper states this directly (p. 596): "In CV auctions, social gains are zero (and therefore smaller than social costs) for all $n \geq 2$." The entry_contest model is therefore in the common-value branch *by construction* — V exogenous and fixed *is* $V_n \equiv V$ — and the benchmark is Proposition 3, with the expected verdict of excessive entry.

The fixed prize is not enough on its own to deliver that verdict. A fixed prize under which an entrant who meets a rival keeps no rent, $W_n = V$ for every $n \geq 2$, also makes free entry optimal, because the entry fee that corrects entry is then zero. The paper's own condition, footnote 13's "as long as W_n varies with n", admits that case when read literally. The excessive-entry verdict therefore needs $V - W_n > 0$ for some $n \geq 2$, and the welfare section must state that entrants beyond the first keep positive expected rent (`lit/levin_smith_1994`, `cv_free_entry_excessive` and `control_no_stealing`).

Their Proposition 9 adds that welfare falls monotonically as the pool N grows beyond n^*, through the coordination cost of mixed entry; Proposition 8 is the supporting statement that the probability of *no* entry rises with N under an optimal mechanism. That is a different mechanism from P7's cap, which is a count bound with no coordination content, but both say that a larger pool does not help beyond a point.

Their equation (9), $(1 - q^*)^{N-1}V = c$, requires care. The page introduces it with "e* must induce entry such that at q*, $\partial S/\partial q$ vanishes" (p. 590), so it is the planner's condition imposed on the equilibrium that the optimal fee induces. It is *not* an entry-equilibrium condition: it is the first-order condition of the welfare problem, obtained by differentiating their equation (8), $S = [1 - (1 - q)^N]V - qNc$, and setting $\partial S/\partial q = 0$. It characterises the socially optimal entry probability, which the seller's optimal entry fee e^* is chosen to induce; the paper's own proof of Proposition 8 writes it as $(1 - q_N^s)^{N-1} = c/V$, with s for social. The bidders' equilibrium condition is their equation (6), $B_i(q^*, \Omega) = 0$. Equation (9) is nonetheless algebraically identical to Fu et al. (2015)'s Definition 1, which reaches the same equation from the opposite direction — as a *lower bound* on equilibrium entry, derived from the win-only-if-alone deviation payoff. The coincidence is worth stating, but the two must not be identified: in common-value auctions the free-entry equilibrium lies *above* q^s (Proposition 3), and reading (9) as an equilibrium condition erases the very wedge that Proposition 3 is about.

Fu–Jiao–Lu, and the accounting behind P7. $M \geq 2$ identical risk-neutral potential bidders, common fixed entry cost, generalised nested lottery contest, prize purse capped at V and allowed to be contingent on the realised number of entrants. Their equation (2) is a rent-accounting inequality: expected prize mass paid out is at least expected entry costs plus expected bidding costs, so

$$[1 - (1 - q)^M]\, V \;\geq\; Mq\,(\Delta + \mathbb{E}[x^\alpha]),$$

which implies that the expected number of entrants Mq cannot exceed V/Δ, since the bracket is at most 1 and $\mathbb{E}[x^\alpha] \geq 0$. Cite *Definition 2 alongside equation (2)* for this: Fu et al. never display the count bound, and what they derive from (2) is a chain about *bids* — (3), then (4) by convexity, (5), the bid bound (6), and Theorem 2's single-peakedness. The count bound is instead the content of their Definition 2, $\bar{q} = \arg\max\{q \in [0, 1] : (1-(1-q)^M)V \geq Mq\Delta\}$, every feasible point of which satisfies $Mq\Delta \leq V$. A referee sent to equation (2) alone will find only the bid chain and conclude the attribution is loose. That is the same accounting logic as P7. P7's content is therefore not a new inequality but a version of a known one for heterogeneous agents, with κ in utility units in place of a nominal cost. Against Fu–Jiao–Lu it also bounds a pure-strategy count rather than an expected count. Against Fu and Lu (2010) neither the

pure-strategy count nor the absence of a designer separates it, as the next paragraph records. The paper should say which distinction applies to which source. What Fu et al. build on top — the uniform upper bound on expected overall *bids* (Theorem 2), its single-peakedness in q, its implementation by Tullock contests with entry fees (Theorem 7), and the optimal shortlisting rule (Lemma 6, Theorem 9, with $\bar{M} = \min\{N : V/N < \alpha\Delta/(\alpha-1)\}$) — are design results with an effort-maximising organiser and have no counterpart in the entry_contest model, which has no designer and a single fixed prize. Their Theorem 11, that the optimal entry probability, the bid bound and the shortlist size all fall in the entry cost, is the design-side cousin of P8.

Fu–Lu. $M \geq 3$ identical, sequential entry with full observation of current participants, organiser chooses prize and a per-entrant fee or subsidy. Their equation (7) states that in the optimally designed contest total effort equals the budget Γ_0 minus N times the entry cost (p. 10), and it drives their Theorem 1, that the optimal contest attracts exactly two entrants. Read as holding at every contest, (7) is false. It holds exactly when the contest breaks even and exhausts the budget, their Lemmas 4 and 5. The inequality $E \leq \Gamma_0 - NC$, and with it $NC \leq \Gamma_0$, does hold at every feasible contest in their model, at its Lemma 2 count (pp. 6–7), although they never state it. "Fixed rules with no designer" therefore separates P7 from their displayed equation and not from their model. Their count is also a pure-strategy, deterministic one (footnote 6, "pure strategies (0 or 1 probability of entry)"; Lemma 2). What separates P7 from Fu–Lu is heterogeneous agents against identical ones, and κ in utility units against a money cost borne by risk-neutral agents (`lit/fu_lu_2010`, `fixed_rules_cap`). Their conclusion lists different types of contestants as future research. So the entire design literature on endogenous entry — Fu–Lu 2010, Fu–Jiao–Lu 2015 — is homogeneous-agent by construction.

Morgan–Orzen–Sefton, and what the experiment did not vary. The theory section gives $n^* = \lfloor\sqrt{P/F}\rfloor$ for identical agents with equal endowment w, outside option F and prize P, under $P > F > P/N^2$ and with $\sqrt{P/F}$ not an integer (p. 441). The design satisfies both conditions, and without them the formula fails (at $P = 40$, $F = 10$ two entrant counts are equilibria). The experiment holds $w = 100$ for every subject by design. Predicted entry was 2 (small prize) and 4 (large prize); observed was 2.5 and 3.7 in the second half (rounds 26–50; 2.7 and 3.6 in rounds 1–25). The *signs differ*: the small prize shows *excess* entry (2.5 against 2, significant in both halves, $p = 0.004$ and 0.012) while the large prize shows a *shortfall* (3.7 against 4, differing from Nash only at the 6% and 12.5% levels). Both deviations run against what within-contest behaviour would predict — with negative returns to two-person competition one expects *fewer* than two, and with under-investment one expects *more* than four — and the paper says of each case "we observe precisely the opposite". Citing this as "observed entry exceeds prediction" would be wrong for half the experiment. Earnings were *not* equalised across the contest and the outside option in either treatment, and a self-selection-by-risk-attitude explanation is rejected by comparing three-player contests across treatments (small-prize triples over-invest and earn less than the outside option, large-prize triples under-invest and earn more; the earnings difference is significant at 1%). The same controlled comparison also defeats a "contest premium" explanation, so two behavioural accounts are ruled out, not one. Two things follow. First, the experimental literature on contest entry has not tested resource sorting, because endowments were held equal; a Morgan–Orzen–Sefton design with heterogeneous endowments is the natural test of P8–P9. Second, their concluding observation that the marginal entrant's power is blunted because inframarginal rent-seeking moves the marginal calculus has no analogue in the binary model, where inframarginal behaviour is fixed at "invest"; that is why the cutoff is clean here and noisy there.

Moreno–Wooders, read in full, and the relabelling objection answered. Moreno and Wooders (2011) has now been read in its published form **[F]**. N risk-neutral buyers in a standard

independent-private-value auction; each buyer's entry cost Z_i is an independent draw from H on $[\underline{c}, \bar{c}]$ and is *privately* observed before entry; the value is learned after entry; the seller may set a screening value v and an admission fee ϕ. Their Proposition 2 gives a unique symmetric equilibrium in threshold form, enter iff $z < t^*(v, \phi)$. When the equilibrium is interior, t^* solves $U(v, H(t)) = t + \phi$ and is decreasing in both instruments (p. 320). The number of bidders is Binomial$(N, H(t^*))$. Proposition 3 shows that $v = \phi = 0$ attains the *constrained* maximum of social surplus, "a constrained maximum in the sense that buyers enter independently according to a symmetric entry rule" (p. 320). The reason given is that the social contribution of a marginal increase in the entry threshold "is proportional to a buyer's utility to entering" (p. 315). The paper's own Table 1 (p. 325) beats that maximum with an entry cap, social surplus .13145 against .10714. Proposition 4: inframarginal entrants keep information rents, so the seller no longer captures the whole surplus. Propositions 5–8: the revenue-maximising screening value is positive but below the fixed-n level; with an admission fee available, screen by entry cost rather than by value; an entry cap plus fee does better still. Proposition 9: heterogeneity is irrelevant asymptotically — only the lower bound $\underline{c}$ matters. Section 6: revenue may rise *or* fall in N, unlike Levin–Smith. The paper signs no effect of a spread of H at finite N. It does compare cost distributions in three other ways, asymptotic invariance to changes that keep the lower bound of the support (p. 326), a two-distribution numerical example (p. 325), and homogeneous against heterogeneous costs (p. 333).

The relabelling objection — is $\kappa \circ \Lambda^{-1}$ substantively different from a primitive cost distribution? — can now be answered on four points, and the first exposes a gap in `PROOFS.tex`.

Information. Moreno–Wooders' costs are private and they study symmetric equilibria, in which every buyer compares her own cost with one common threshold, so only the *distribution* of the entrant count is pinned. The common threshold comes from the symmetry of the equilibrium they study, rather than from private information alone (p. 320). `PROOFS.tex` writes the entry condition as $\kappa(w_{(j)}) \leq \Delta(j-1)$ on the order statistics of the wealth profile, which presupposes that the profile is common knowledge, and it now states that assumption (§Primitives, Information). Common knowledge makes k^* deterministic and the same in every pure-strategy equilibrium. It does not pin the identities of the entrants, which an earlier version of this survey said it did. It is the first structural difference from Moreno–Wooders and the reason P5 delivers an exact count where they deliver a binomial.

Primitive. Their H is a primitive and they never spread it at fixed N. Ours is induced from a resource distribution through a known concave map, so a mean-preserving spread of Λ is a specific, non-arbitrary deformation of the induced cost distribution. P9 has no counterpart in Moreno–Wooders because there is no distributional comparative static there to have a counterpart to.

Designer. Theirs is mechanism design with three instruments; ours has none.

Objective. Their Proposition 3 is the independent-private-value logic, in which the marginal entrant's private gain is proportional to her social contribution, and its optimum is constrained to symmetric independent entry rules. Under complete information k^* is not restricted to such rules, so the welfare section must name the benchmark it compares against. For a fixed-prize winner-take-all contest the common-value logic of Levin and Smith (1994)'s Proposition 3 applies instead, and free entry is excessive provided entrants beyond the first keep positive expected rent. Their Proposition 4 carries over directly: entrants with $w > \bar{w}$ keep surplus.

So the threshold-in-cost-space is theirs; the count-pinning under complete information, the induced-cost structure, and the distributional comparative static are not. The objection is answered, and the information assumption it needed is now written down in `PROOFS.tex`.

Still at abstract level. Kaplan and Sela (2010) give a rationale for entry fees in an all-pay auction with privately known entry costs. **[C]** Ginzburg (2021) has a continuum of contestants and a profit-maximising entry price. **[A]** Grossmann and Dietl (2015) has high- and low-talent contestants choosing between entry and heterogeneous outside options. **[A]** Samuelson (1985) is the interim-entry-cost variant (cost incurred after learning one's value) that both Levin–Smith and Moreno–Wooders position against. **[C]**

5 Budget and liquidity constraints

Che and Gale (1997) has now been read in full **[F]**, in its published form. The setting is $N \geq 2$ risk-neutral bidders with common value v, publicly known wealths $w_1 > \cdots > w_N$, and a hard cap: a bidder cannot spend more than his wealth (no borrowing in §3; credit at rate r in §4). Utility is linear; wealth enters only as the cap. There is no entry cost and no participation decision.

Their Proposition 1 (all-pay auction) gives expected total expenditure $\min\{v, w_2\}$ with only the two wealthiest active — the rich preempt the poor. Their Lemmas 1–3 (lottery) give a unique equilibrium with a critical bidder i^*: those with wealth above a common bid b^* all bid b^*, those below bid their wealth, and everyone participates. Proposition 3: the lottery dissipates more than the all-pay auction iff v exceeds a threshold above w_2. Adding low-wealth entrants has no effect in the all-pay auction and always raises lottery expenditure.

This reading confirms first-hand what §2 previously took from Schroyen–Treich's description: a budget constraint is a kink in an otherwise linear payoff at w_i, and nothing else. The entry_contest model has no kink and no cap; $\kappa(w)$ is smooth and the sorting comes from curvature everywhere. The critical-bidder structure in their lottery is the hard-constraint cousin of the κ threshold, but it operates on the intensive margin (bid size) while ours operates on the extensive margin (enter or not), and in their lottery participation is never a choice. Their all-pay cap $\min\{v, w_2\}$ is wealth-driven and depends on the second-highest endowment; P7's cap V/κ is prize-driven and independent of the distribution. Their observation that adding low-wealth agents leaves all-pay expenditure unchanged rhymes with anonymity plus P7 — agents who would not enter do not move k^* — and is worth a sentence.

Grossmann and Dietl (2012) consider two bidders with asymmetric valuations under imperfect capital markets and find aggregate investment lower when at least one is constrained. **[A]** Grossmann and Hottiger (2020) and Che and Gale (1998) continue the line. **[C]**

6 Risk aversion and concavity in contests, read in full

Cornes and Hartley (2012) has been read in full as Manchester Economics Discussion Paper EDP-0806 (May 2008), the working version of the *Economic Theory* article. **[F, WP version.]** The setting is $n \geq 2$ contestants with logistic CSF and concave production, an indivisible common rent R, continuous expenditure, concave utility u_i, complete information; no entry cost.

The multiplicity example is now concrete. Their Example 1 has ten identical contestants with $u(c) = c - 0.45c^2$ — increasing absolute risk aversion — and possesses both the symmetric equilibrium with each spending 0.0563 and an asymmetric one in which three contestants spend 0.184 and seven spend zero; the active three earn more than in the symmetric equilibrium. Uniqueness (Theorem 4.2) needs *regularity*, and their sufficient condition (Lemma 4.2) is a curvature bound $2u'(R - x) \geq u'(-x)$ on $(0, R)$; it holds for CRRA when initial wealth $I_i \geq R(1 - 2^{-1/\gamma})^{-1}$ (Corollary 4.1), for small rents (Corollary 4.2), for prudent contestants with

absolute risk aversion at most $1/(2R)$ (Corollary 4.3), and for CARA. Whether non-increasing absolute risk aversion suffices is left open.

The contrast with P5 is now exact. Their asymmetric equilibrium is a coordination among *identical* agents on who is active, sustainable because the continuous aggregative best response admits several fixed points under IARA. P5 has agents ordered by $\kappa(w)$, so who is active is pinned by the order and there is nothing to coordinate on; P5 needs only that κ is decreasing, with no curvature bound of the Lemma 4.2 type. That is the reason the binary-entry structure escapes their multiplicity, and it should be stated in one sentence with their Example 1 as the foil.

Two further results are directly relevant. Their dropout value $\bar{Y}_i = f_i'(0)[u_i(R) - u_i(0)]/u_i'(0)$ (Lemma 3.2) makes a contestant inactive whenever aggregate lobbying exceeds it — a curvature threshold on the *aggregate*, where $\kappa(w)$ is a curvature threshold on *own* resources. And Proposition 6.2 is a selection result: in large asymmetric contests only the types with the highest dropout value stay active, so competition drives the more risk-averse out and the limiting dissipation ratio $H[u] = [u(R) - u(0)]/(Ru'(0))$ is that of the least risk-averse type. That is selection by risk attitude across heterogeneous u_i with common wealth; ours is selection by resource level with common u. The two are complementary and should be cited together. Their CRRA specification $u(c) = (I_i + c)^{1-\gamma}/(1-\gamma)$ has initial wealth as an additive shift in a static game, the same one-period object as in Schroyen–Treich.

Treich (2010) has been read as LERNA working paper 09-013 of 17 February 2009 [**F, WP version**]. $n \geq 2$ identical agents with initial wealth w, concave u, rent b, payoff $p_i u(w+b-x_i) + (1-p_i)u(w-x_i)$, general CSF. Proposition 2: prudence ($u''' \geq 0$) is sufficient for risk aversion to lower effort relative to risk neutrality, for any n and any regular CSF; Corollary 1: for $n = 2$ it is necessary and sufficient, and quadratic u gives no effect at all. Proposition 1: DARA plus a small rent gives a unique symmetric equilibrium. The paper's closing sentence lists entry as future work. Two things follow. The risk-aversion strand needs u''' to sign anything, whereas P8 and P9-gen need only u''; that is a concrete statement of what "no higher-order restrictions" buys. And wealth here is again a one-period additive shift. Skaperdas and Gan (1995) is the antecedent; Cornes and Hartley (2005) is the general-technology companion; Cornes and Hartley (2003) establishes regularity under CARA. [**C**]

7 Competitor numbers and the shape of noise, read in full

Ryvkin and Drugov (2020) has been read in its published *Theoretical Economics* form [**F**], and Drugov and Ryvkin (2020a) as the working paper of 13 March 2019 [**F, WP version**].

The setting is symmetric risk-neutral players, output $y_i = e_i + X_i$ with i.i.d. additive noise, a winner-take-all prize normalised to one, strictly convex cost, and a number of players K that may be deterministic or stochastic; heterogeneity is assumed away explicitly (their footnote 3). There is no entry margin. The two objects that drive everything are

$$b_k = \mathbb{E}\big[f(X_{(k-1:k-1)})\big], \qquad E_k^* = k\,c'^{-1}(b_k) = \mathbb{E}\big[h(X_{(k-1:k)})\big] \text{ under quadratic cost,}$$

the density of noise at the largest of the other $k-1$ draws, and the hazard rate at the second-largest of k. Proposition 1 and Corollary 1: a unimodal noise density is inherited by individual effort as a function of k, by a Karlin variation-diminishing argument on $F^{k-1}(1-F)$. Corollary 4 gives monotone effort for a monotone density, constant iff uniform, and its parts (ii) and (iii) carry the condition $p_1(\theta) = 0$ (p. 1600). For a deterministic number of players, $e_2^* = e_3^*$ with a decrease thereafter under a symmetric unimodal density is credited by Ryvkin and Drugov to Gerchak and He (2003) (p. 1600), and their Proposition 2 generalises it to a stochastic number of players. Proposition 3: IFR noise with cost more convex than quadratic gives aggregate

effort increasing in K; DFR noise with cost less convex than quadratic gives aggregate effort decreasing. Proposition 5: for a mean-preserving increase in uncertainty about K, log-concave noise lowers aggregate effort and log-convex raises it. Proposition 7 (Appendix A.1) gives existence and uniqueness of the symmetric pure-strategy equilibrium under a lower bound on c'' and bounds on f and f'.

The relationship to P-MU can now be stated at the level of objects rather than by analogy. P-MU's $\Delta(0)$ is the win-probability gain to the first investor, a functional of the induced score distributions; its dependence on Q runs through the same order-statistic weighting that gives b_k its dependence on k. Proposition 1's inheritance of unimodality is the structural template for P-MU's hump-shaped weighting, and the sentence on their p. 1589 — that no prediction about the effect of competitor numbers survives for every noise distribution — is exactly the frame in which R2's Q-conjecture should be refuted. The comparison of P-MU's weighting with their kernel has since been computed, and its result, including a correction to an earlier version of this survey, is item 3 of the verdict below. Their footnote 29 records that Drugov and Ryvkin (2020b) shows the dispersive order is necessary and sufficient for ranking equilibrium effort across noise distributions — the dispersive order again, now in the contest branch as well as the status branch.

The 2019 working paper is the prize-allocation companion. With prizes summing to one and monotone in rank, the optimum is a two-prize schedule with the top r^* ranks paid equally (Proposition 1), where r^* maximises $B_{r,n}/r = \frac{1}{n}\mathbb{E}[h(Z_{(n-r:n)})]$; IFR gives winner-take-all, DFR gives reward-all-but-last (Proposition 2); the convex transform order gives monotone comparative statics in skewness (Proposition 4); log-concave noise makes effort Schur-convex in the prize vector (Proposition 7). Proposition 8 applies this to status tournaments in the sense of Moldovanu et al. (2007): with IFR noise the top status category is a singleton, with DFR it contains $n-1$. This is not P-MU territory — the entry_contest model has one prize — but it is the status-tournament link back to the abandoned paper's motivation, and a referee from that literature will expect it cited. Lim and Matros (2009) is the binomial-K special case. **[C]**

8 Assignment models and the pivot rule, read in full

Two papers previously listed as "assignment-model comparators" turn out to carry the closest published precedent for P9-gen's pivot rule. Costrell and Loury (2004) has been read as the draft of 11 December 2003 and, on 13 October 2026, in its published form **[F]**, with the same proposition numbers and statements (record in `lit/costrell_loury_2004`); Suen (2007) has been read in its published form **[F]**.

Costrell–Loury and the single-pivot precedent. In their two-job model (§2), jobs are filled in proportions θ and $1-\theta$, and the entire wage schedule is determined by one sufficient statistic: $\hat{\mu} = F^{-1}(\theta)$, the ability of the *marginal* worker between jobs. Any shift that raises $\hat{\mu}$ raises low-ability wages and lowers high-ability wages. Under a mean-preserving spread on fixed support, they state that the effect depends on whether the marginal worker at quantile θ lies in the upper or lower tail: if θ is high the spread raises $\hat{\mu}$ and narrows the wage distribution, if θ is low it lowers $\hat{\mu}$ and widens it. The rule reads which tail the marginal agent's quantile lies in, and it allows "possible multiple crossings in between" (p. 9). For a single-crossing spread that is a single-pivot rule at the crossing point, the same logic as P9-gen, where the sign of $\mathrm{d}k^*$ flips according to whether $w_{(k^*)}$ lies above or below x_0. Their object is a wage schedule through $\hat{\mu}$; ours is an entry count through $w_{(k^*)}$. P9-gen should be presented as the entry-count analogue of the Costrell–Loury two-job comparative static. Their continuum result (Proposition 6) then shows what governs the direction when there is no single marginal agent: with G riskier than F on $[0,1]$, concave β lowers $w(0)$ and widens the span, convex β lowers $w(1)$ and narrows it.

The curvature of the payoff map decides. For us that map is κ.

Three further results are tools. In the proof of their Lemma 1, which serves Proposition 6 (p. 20), they assert that G riskier than F on fixed support implies the quantile functions reverse the order, F^{-1} riskier than G^{-1}, and refer to the proof of Proposition 5 for it; their Proposition 5 signs an aggregate (output) under a general spread with multiple crossings by integrating $\Gamma(\rho) = \int_\rho^1 [G^{-1} - F^{-1}]$ against $\mathrm{d}\beta$; and their Theorem 1 shows that a Blackwell-more-informative signal induces a second-order-riskier distribution of posterior means, so any mean-preserving-spread result has an information reading. Their Proposition 10 (Cobb–Douglas crowding) removes the curvature dependence, since a spread there always narrows the span. It therefore reverses Proposition 6 only where Proposition 6 widens the span, that is "for concave $\beta(\cdot)$" (p. 28), and under convex β both narrow it. Any statement that re-optimising the assignment reverses the inequality effect has to carry that condition. The draft is not the published article; the published version should be checked for these proposition numbers.

Suen and the general-spread technique. One-to-one transferable- utility matching with $Q = \theta(x)\phi(y)$, positive assortative matching, and matching function $\mu = G^{-1} \circ F$, which they note can itself be read as a distribution function. Proposition 2: if G_1 is more dispersed than G_0 (same mean, second-order), then total earnings on the other side fall, under concavity of θ, ϕ and log-concavity of $1 - F$. The proof handles a *general* mean-preserving spread, in Suen's own definition (p. 153, following Mas-Colell, Whinston and Green), with an arbitrary finite number of crossings $k_0 < k_1 < \cdots < k_{2n}$ of the quantile functions: the quantile difference alternates in sign, each partial integral is signed by second-order dominance, concavity controls the recombination, and the result is integrated by parts against $\rho = (1 - F)/f$. Proposition 3 and its corollaries give conditions under which every agent loses. Suen's μ is cross-side (the other market's distribution moves) whereas our $T = \Lambda_\beta^{-1} \circ \Lambda_\alpha$ is same-side (before and after), but the map class is identical.

Consequence for open item 2. Two published sources sign an aggregate under a general mean-preserving spread, by the same first move but under *different* hypotheses — and they are not independent: Suen's proof imports the quantile-reversal step from Costrell–Loury by citation ("see, for example, [2]"), and his footnote 1 engages their rank-rescaling device directly. One lineage, not two convergent confirmations.

Both begin by decomposing the quantile difference at the crossing points. They then diverge:

- Costrell and Loury (2004)'s Proposition 5 integrates $\Gamma(\rho) = \int_\rho^1 [G^{-1} - F^{-1}]$ against $\mathrm{d}\beta$ and needs only that β is **non-decreasing**. It requires no concavity and no hazard-rate condition; the strings "log-concav" and "hazard" do not occur anywhere in that paper. Concavity of β belongs to their Proposition 6, a different result about wage *inequality*.
- Suen (2007)'s Proposition 2 integrates by parts against $\rho = (1 - F)/f$ and assumes concavity of θ, ϕ, which Suen calls "sufficient but not necessary" (p. 155), *and* log-concavity of $1 - F$, the latter being exactly equivalent to $\rho' \leq 0$, since $\mathrm{d}^2 \log(1 - F)/\mathrm{d}x^2 = \rho'/\rho^2$. Concavity cannot simply be dropped, since a convex ϕ with every other hypothesis holding reverses the conclusion (`lit/suen_2007`, SU-7).

The route was not needed. `PROOFS.tex` closes open item 2 through the margin condition (Proposition (margin condition) and its band), rather than through this machinery. Both sources sign a smooth aggregate and so must control the quantile difference at every rank, whereas k^* is a threshold count that reads the quantile difference at two ranks, the marginal entrant and the first outsider. The three obstacles an earlier version of this section recorded (aggregate against count, the rescaling trap of Suen's footnote 1, cross-side against same-side)

therefore do not arise. If an aggregate version is wanted later, the Costrell–Loury monotone-weight route is the weaker one and the place to start.

9 Positional competition and education

Chung and Lee (2017) is the nearest thing to the abandoned paper's framing: inequalities in wages, productivity and education affordability determining educational effort in competition for premier jobs, in a Tullock framework with endogenous returns, finding that competition for better-paying jobs can produce socially inefficient excessive effort and that a widening wage gap intensifies the competition. **[A]** Two players, continuous effort.

10 Foundations for a rank-allocated prize, read in full

Three papers previously cited secondhand have now been read in full, and they do a specific job: they justify the premise that `PROOFS.tex` currently asserts, that a fixed prize V is allocated by rank rather than by price.

Cole–Mailath–Postlewaite 1992. Published form **[F]**. Multigenerational; a continuum of men with capital and women with a non-traded endowment; joint consumption; CRRA. Status is defined as a ranking device that determines how one fares in the non-market sector. Two social norms: wealth-is-status and aristocratic (inherited). Proposition 1: a wealth-is-status equilibrium exists (Mas-Colell's continuum-game theorem). Property 1: in equilibrium no family's relative rank ever changes, even though everyone competes. Property 2 says every family weakly over-saves relative to the no-matching benchmark, strictly for all but a set of families of measure zero, and the family matched with the zero-endowment woman is undistorted (p. 1107). Proposition 2 gives an aristocratic equilibrium when $\gamma > 1$, $k_0(0) = 0$, β is close enough to one, and capital is sufficiently "spread out" at the lower tail (p. 1111). Their footnote 20 says a market in which relative position determines the house one buys *cannot* generate the multiplicity they are after — the prizes must be non-market.

Their §IV.A two-period example compares two men "with the same initial income level in two different economies", and concludes that the one in the economy with the more compact income distribution "will tend to have a higher savings rate, all other things being equal" (p. 1103). The comparison covers the top half of the distribution, and it is not mean-preserving, since the compact economy sits at the top-half capital level $K + \varepsilon$ (pp. 1102–1103). §V.D adds that "in the wealth-is-status savings model with $\gamma < 1$" the incentive to save disappears once the distribution is dispersed enough (p. 1117). That is the 1992 origin of the "standard result" R2 invoked, on the intensive margin.

The channel matters for how the entry model may cite it. In the example a man's own income is held fixed and his savings fall because his rivals' wealth is more dispersed, through the matching. Proposition (anonymity) in `PROOFS.tex` excludes exactly that channel from the entry model, where a challenger's gain from entering depends on the number of rival entrants and not on their wealth. The entry model therefore shares the direction of the standard result and not its mechanism, and a claim that the sign rule "reconciles" the two can rest on the direction only.

Cole–Mailath–Postlewaite 1995. CARESS Working Paper 95-14, 1 August 1995, marked forthcoming in the Minneapolis Fed *Quarterly Review* **[F, WP version]**. The working-paper title has "Concern" singular; Hopkins–Kornienko's 2010 reference list has "Concerns" plural at *QR* 19(3), 12–21. The published title has not been confirmed and the bib entry carries the working-paper form. The concern for relative standing is *instrumental*: rank matters only because a non-market decision depends on it. §2 is a complete-information effort model with the

matching function $m(y)$ equal to the distribution function of output, so m' is large when the distribution is tight and competition is intense (§2, p. 5). §3 is the incomplete-information version: wealth is unobservable, so the rich signal by destroying wealth, and the fraction destroyed rises with rank. In the body of §2.1 (p. 7), which invokes footnote 8, it is competition from *below* that distorts effort. Truncating the population from the top changes nothing, and truncating it from the bottom leaves the new lowest agent undistorted. Footnote 8 itself (pp. 5–6) concerns the undistorted lowest-ability female. §4: when prizes are allocated by rank rather than sold, the welfare theorems do not apply.

What these two give the entry_contest model. Finding 5 asks why threshold entry into a rank-allocated prize is not already covered. The answer to the "rank-allocated prize" half is that Cole–Mailath–Postlewaite is where that premise is justified and where its consequences (no welfare theorems, non-market allocation, instrumental status) are worked out. The manuscript now cites them where V is introduced (`MANUSCRIPT.tex`, Appendix D) and says that V is a prize in their sense, allocated by rank and not sold. Their "competition from below" is the same structural fact as the anonymity proposition and the ordering by κ: only agents who could outrank you matter. Their Property 1 is consistent with the model here — entry does not change the wealth rank; it is the score, not the rank, that the prize follows. And their multiplicity is *across* norms, not within one, so it is not in tension with P5.

Becker–Murphy–Werning. Becker et al. (2005), draft revised December 2003 [**F, draft version**]. Utility $u(c, s)$ with the crucial assumption $u_{cs} > 0$: status raises the marginal utility of consumption. A fixed supply of status positions, sold in a hedonic market (or, equivalently by their Proposition 1, assigned by rank in consumption of a fixed-supply, intrinsically worthless social good). Stage one: fair lotteries over income. Proposition 2: if the constant-marginal-utility allocation is a mean-preserving spread of the initial income distribution, it is the competitive lottery equilibrium — a minimum equilibrium inequality that any more compact initial distribution is spread out to reach. Proposition 3: with a status market this coincides with the utilitarian planner's allocation, so the planner *increases* inequality. §7: when status attaches to income *rank* rather than being purchased, rank lotteries impose real externalities (one's win lowers others' rank) and the equilibrium no longer coincides with the planner's. This is not a contest paper; the mechanism is orthogonal to κ, the mean-preserving spread is an *output* of equilibrium rather than an input, and $u_{cs} > 0$ is the structural opposite of our separable V. It matters for one thing only: the welfare section (R2 Comment 3). Their §7 states in print that rank-allocated status carries real externalities that purchased status does not, which is the zero-sum, business-stealing frame the welfare benchmark needs, and their Proposition 3 is a warning that a planner may *want* inequality.

Chung and Lee (2017), Cole et al. (1998), Postlewaite (1998), Frank (1985) and Bilancini and Boncinelli (2012) remain at abstract or citing-paper level; Fernández and Galí (1999) was read in full on 14 October 2026 (§17, row L30).

11 Refuted conjectures in context

Referee 2 conjectured that for $\mu > 0$, higher Q shrinks the pie among investors so fewer invest. `PROOFS.tex` §R2 refutes this with an exact family of the model in which the first entrant's gain rises strictly with Q at every $\mu \leq \frac{1}{2}$ (`lean/mathlib/Refutations.lean`, `bern_Delta_strictMono`), and with another exact family in which the gain falls with Q (`lean/mathlib/FallWitness.lean`, `fall_from_one`), so the sign is set by the primitives. Placed against Ryvkin and Drugov (2020), this refutation is the expected outcome rather than a surprise, and framing it that way is both more honest and more persuasive than presenting

it as an isolated counterexample.

12 Background and toolkit

Lazear–Rosen, read in full, and a precedent that was hiding in plain sight. Lazear and Rosen (1981), published form [**F**]. The risk-neutral two-player tournament: $C'(\mu) = (W_1 - W_2)g(0)$, efficient, with the prizes in equation (10) equal to expected product plus or minus $V/2g(0)$, which they read as an *entrance fee or bond*: each player posts a bond and takes a fair winner-take-all gamble over the pool. Footnote 2: a pure-strategy equilibrium exists only when the noise variance is large enough — the ancestor of Ryvkin–Drugov's Proposition 7. §III adds risk aversion, and the subsection titled "Income Distributions" contains the result that matters here. With a common utility function and different *endowed* incomes y_0, "persons with more endowed income and smaller absolute risk aversion are more likely to prefer contests", and if the difference is large enough "it can be optimal to pay piece rates to those with small values of endowed income and to pay prizes to those with large values" (p. 855). Their Table 1 shows the sorting at one of the three variances it tabulates for both endowments, $\sigma^2 = 1$. So agents *can self-select into the contest by wealth*, and the resulting income distribution is positively skewed. That is the first appearance, in 1981, of "who enters a contest is determined by wealth". The channel is risk attitude — DARA, so the wealthy find the gamble cheaper — not the utility cost of a fixed fee, and it needs u''' where κ needs only u''. But it is a precedent for the mechanism and it should be cited as one, with the channel distinction stated. §IV gives adverse selection into the top league when each person's class is private (p. 858), and a competitive handicap $h^* = \Delta\mu/2$ when the types are "known to everyone" (p. 861). The handicap algebra, with its public types, is the tool for R2's open question about an equilibrium in which the incumbent abstains and a challenger invests.

Moldovanu–Sela and Hoppe–Moldovanu–Sela, read in full. Moldovanu and Sela (2001) as the draft of 5 June 2000 [**F, draft version**]: $k \geq p$ contestants with *private ability* entering a separable cost $c_i\gamma(x)$, all-pay, perfectly discriminating; a single first prize is optimal under linear or concave cost (Propositions 3.2, 4.2) and two prizes can be optimal under convex cost (Proposition 4.3). Their §5 entry fee produces a threshold in *ability* space — a third threshold variant, alongside Moreno–Wooders' cost space and our resource space. Their footnote 12 warns that independence of types is problematic with endogenous entry, which applies here too: the entrant pool is the top k^* by w, not an i.i.d. sample. Hoppe et al. (2009), published form [**F**]: finite two-sided matching on costly signals with private attributes, built on Barlow and Proschan (1966)'s normalised-spacings results. Proposition 4: entry on the long side raises own-side signalling for every F, and moves the other side's signalling up or down according to whether F is DFR or IFR; Example 1 shows welfare falling in n up to $n = 8$ in a small market and rising thereafter. Proposition 5: the relatively *more homogeneous* side signals more, because it perceives a more dispersed prize structure. That is the perfectly-discriminating, private-ability counterpart of the competitor-number comparative static: with no noise, entry always raises own-side effort; P-MU says that with noise the sign at the margin is distribution-dependent even for symmetric agents. We read their IFR/DFR spacings machinery and Ryvkin–Drugov's hazard-rate machinery as one Barlow–Proschan lineage. The attribution is ours, since Ryvkin and Drugov do not cite Barlow and Proschan and take their orders from Shaked and Shanthikumar (2007).

Surveys and standard references. Konrad (2009) and Corchón (2007) are the two surveys a referee will expect to see acknowledged; the 2010 Hopkins–Kornienko text cites Konrad for the point that greater heterogeneity among competitors and a smaller spread of prizes both reduce equilibrium effort. [**F, as characterised there.**] Moldovanu and Sela (2006) is the

contest-architecture companion and Schotter and Weigelt (1992) the experimental treatment of asymmetric tournaments. **[C]**

Two references carry technical weight rather than positioning weight. Shaked and Shanthikumar (2007) is the source for the dispersive and star orders and is the right citation for the pivot-spread class discussed in §3. Rothschild and Stiglitz (1970) is the definition open item 2 of `PROOFS.tex` appeals to when it observes that a general mean-preserving spread admits only a mean-preserving coupling $w' = w + \varepsilon$ with $\mathbb{E}[\varepsilon \mid w] = 0$, which need not be monotone. Citing both, and stating that the pivot-spread hypothesis is the dispersive-order restriction rather than an arbitrary one, converts that open item from an admitted weakness into a deliberate and named choice of stochastic order.

13 Verdict on novelty

The mechanism alone does not carry a paper, and after the full read of Schroyen and Treich (2016) it carries even less than the previous draft of this document supposed. "Resource-sorted entry via concavity" is their privilege contest plus a Moreno and Wooders (2011) threshold on the entry structure, and the resource-label distinction does not separate the two (§2). Nor is a mean-preserving spread in the resource distribution new in itself: their Theorem 3 already signs one inside a privilege contest.

Two further sharpenings from the full reads. Lazear and Rosen (1981)'s "Income Distributions" subsection (within §III) already has agents self-selecting into a tournament by endowed wealth, through DARA; the mechanism has a 1981 precedent and the contribution sentence must acknowledge it. Two qualifications, both verified, bound how much it concedes.

First, *their claim is an example and they say so.* The subsection opens "While it is not possible to make a general argument based on an example, table 1 suggests…", and rests on one utility ($U = ay^a$), quadratic costs, normal errors and two endowment levels. The honest acknowledgement is that Lazear and Rosen *suggest* the sorting on an example under DARA, while the present model *derives* it under concavity alone, distribution-free.

Second, the channel difference (u'' here, u''' there) is substantive rather than verbal, and separates decisively. Our $\kappa(w) = u(w) - u(w - c)$ is strictly decreasing whenever $u'' < 0$, with no condition on u'''; DARA needs $A'(w) < 0$, whose numerator is $(u'')^2 - u'''u'$, so it *requires* $u''' > 0$. Quadratic utility witnesses the gap: it is concave, has $u''' = 0$, and is *increasing* absolute risk aversion — our sorting operates on it, theirs cannot.

A third difference, which the channel formulation understates, is the *object of choice*: theirs is a choice between two compensation schemes that both pay, so the sorting is tolerance for the binomial spread of a lottery; ours is whether to pay a fixed fee at all, so the sorting is affordability in utility units. That is different economics, not merely a different derivative, and it is the more robust half of the answer.

And Moreno and Wooders (2011) (§4) shows that what separates P5 from a private-cost threshold is the complete-information assumption on the wealth profile, which `PROOFS.tex` now states.

What is not in any source located here is the combination of three results that require the Q-player binary-entry setting jointly:

1. **P7**, the uniform cap $k^* \leq V/\kappa$ holding for every Q with no limit taken. The accounting logic behind it — entrants times entry cost cannot exceed prize mass — is already in Fu et al. (2015), equation (2) *together with their Definition 2* (they never display the count bound itself; see §4), and Fu and Lu (2010), equation (7), for identical agents with mixed or sequential entry. Fu–Lu's (7) is an equality for total effort in the optimally designed

contest, and as a bound on the count it is slack there by $\Gamma_0 - 2C$. Their model also yields $NC \leq \Gamma_0$ at every feasible contest, without a designer, and their count is pure-strategy and deterministic (§4). P7's content is therefore the version of this accounting for heterogeneous agents in utility units, and against Fu–Jiao–Lu alone also a bound on a pure-strategy count rather than an expected one. It should be presented as that, not as a new inequality, and the novelty ledger in `TODO.md` records it as not novel.

2. **P9 and P9-gen**, the pivot rule locating the sign reversal of the inequality effect on aggregate entry at $w_{(k^*)}$ versus x_0. The novelty is not that a mean-preserving spread is signed — Schroyen and Treich (2016) do that — but that it is signed *globally and without higher-order restrictions on* u, where the intensive-margin result is local and needs the CSF decisiveness parameter and the third derivative. The pair that exhibits the third-derivative dependence, CARA against log utility, separates only for $m < 2^{-1/2}$ (§2). The literature supplies the stochastic order (dispersive) but not the entry-count application. The closest structural precedent for the pivot rule itself is Costrell and Loury (2004)'s two-job model (§8), where the sign of a spread's effect depends on whether the marginal worker's quantile lies in the upper or the lower tail; P9-gen is its entry-count analogue and should cite it as such.

3. **P-MU**, the exact condition for the sign of the competitor-number effect in a binary-entry contest, as the discrete analogue of the Ryvkin and Drugov (2020) *density* result $b_k = \mathbb{E}[f(X_{(k-1:k-1)})]$ — *not* their hazard-rate result, which governs *aggregate* effort as $\mathbb{E}[h(X_{(k-1:k)})]$ and uses a different order statistic (the two densities stand in ratio $k(1-F)$). Since $\Delta(0)$ is an individual gain, b_k is the right comparator. Note that Schroyen and Treich (2016) list an arbitrary number of players as an extension they did not undertake, so the Q-dimension is unoccupied on both sides.

   The comparison has now been computed. P-MU's weight is $W = G^{Q-1}(1-G)F$, and its factor $G^{Q-1}(1-G)$ is Ryvkin and Drugov's log-supermodular kernel $-H_\theta = F^{k-1}(1-F)$ under $G \leftrightarrow F$, $Q \leftrightarrow k$; our extra factor F is the incumbent's CDF and carries no Q. The kernel is theirs. Its peak at $(k-1)/k$ is our computation, which they do not state and their argument does not use. Log-supermodularity transfers (the cross-partial of $\log[z^{n-1}(1-z)]$ is $1/z > 0$).

   The single-crossing hypothesis transfers too. An earlier version of this item said it did not, because it dropped the leading minus sign of the P-MU identity. Since $f - g = -\varphi'$, that identity reads $\Delta(0, Q+1) - \Delta(0, Q) = V \int G^{Q-1}(1-G)\,(F\varphi')\,\mathrm{d}x$, so the function in the role of Karlin's u' is $F\varphi'$, which crosses $+-$ whenever φ is single-peaked, the orientation their argument needs. With that premise the difference changes sign at most once in Q, from positive to negative, so $\Delta(0, Q)$ is monotone in Q or rises and then falls, and an interior *minimum* is excluded. An exact example with $F = x^4$ and $G = 1-(1-x)^5$ rises and then falls, the difference being $1/9009$ and $106/2909907$ at $Q = 1, 2$ and negative for $Q = 3$ to 6 (`lit/ryvkin_drugov_2020`, RD-8, recomputed independently). The continuous variation-diminishing step is proved on measures, without densities (`lean/mathlib/PMU.lean`, `pmu_single_crossing`, `pmu_quasiconcave`), and φ is single-peaked for the model's induced F and G when r is uniform (`pmu_quasiconcave_uniform`). Other laws of r are open. The author has adopted the rise-then-fall result as a claim of the paper, and its novelty against Ryvkin and Drugov (2020) is assessed with the other novelty-bearing readings (`TODO.md` item L3).

An earlier version of this section concluded that the contribution should be stated as those three theorems. The novelty ledger in `TODO.md` has since narrowed it. P9-gen survives, with its endogenous margin and its margin condition, as the claim that the sign of the inequality effect on entry is set by the marginal entrant's position against the pivot under burden-monotonicity

alone, for which $u'' < 0$ suffices. P7 does not survive as novel, because its accounting is Fu–Jiao–Lu's and Fu–Lu's. P-MU does not survive as such, because its kernel and its message that no competitor-number prediction is universal are Ryvkin and Drugov's. Four papers that bear on this verdict, one of them on P5 directly, were read after it (§15).

14 Sources read for the measurement map

Two sources were read in full [**F**] for `MEASUREMENT_MAP.tex`, the companion note that says how each object of the model could be observed. Neither bears on novelty, and neither is cited in `PROOFS.tex`.

Schoemaker (1982) is the reference for reading the model's utility as a von Neumann–Morgenstern index. Such a utility "is used to represent preferences" rather than determining them (p. 532), it is unique up to $aU + b$ with $a > 0$ (p. 531), and as a preference theory it "provides no more than ordinal rankings of lotteries" (p. 533). The entry condition is a comparison of expected payoffs (`checks/verify_measurement_map.py`, MM9), so the model is of that kind, and κ and V carry meaning only through κ/V.

Sen (1973) supplies the distinction between preference as "the binary representation of individual choice" (p. 253) and preference read as welfare, of which "it is not in general possible to guarantee both simultaneously" (p. 259). The distinction bears on the unweighted sum of payoffs by which the assortative selection ranks equilibria, and on the welfare section still to be written. Sen (1971), which Sen (1973) cites for the conditions on choice functions, has not been read.

15 Leads read after the verdict

Morgan et al. (2012) (p. 442) place their model against four papers, none of which this survey had cited. Each bears on the novelty verdict, and all four have now been read.

- Fullerton and McAfee (1999) "study contests where agents are heterogeneous and entry is determined by an auction". Read in full since, [**F**]. The auction and the efficient entry equilibrium both select entrants by type, and the consequences for P5 are stated in §4.
- Mathews and Namoro (2008) "consider entry decisions when each agent faces multiple possible contests", in rank-order tournaments with endogenous entry. Read in full since, [**F**], as the working paper of 29 March 2005; the published version is not checked. Each of a strong and a weak contestant joins one of two events, and the strong one can join the event with the smaller prize. There is no entry cost, the number of strategic contestants is fixed at two, and wealth does not appear. Its record is in `lit/mathews_namoro_2008` and its Lean file is `MathewsNamoro.lean`; the row is L23 of `MANUSCRIPT.tex`. Three misprints in the working paper are recorded there (pp. 9 and 11).
- Corcoran (1984) gives the long-run rent-seeking model with free entry of which Morgan, Orzen and Sefton call theirs "a slightly modified version". Read in full since, [**F**]. The players are identical and their number is a real number; the long-run count is $r/(r-1)$ and total expenditure equals the discounted prize. Its record is in `lit/corcoran_1984` and its Lean file is `Corcoran.lean`; the row is L24 of `MANUSCRIPT.tex`.
- Corcoran and Karels (1985), the companion paper, answers Tullock's objections to the long-run count. Read in full since, [**P**], from the journal pages, which are page images; its quotations are checked against an OCR text and its equations and tables against the images. The players are identical and bet continuously, and the count is set by the exponent r, by the commitment assumed (pre-emptive bets against hit-and-run or

hardball entry) or by a minimum bet. With a minimum bet Z, which the paper reads as a fixed entry fee, the count is $\lfloor X/Z \rfloor$, M10's bound binding, because those who do not bet cannot win. Under hit-and-run entry several counts are stable, because incumbents choose among pre-emptive bets. Its record is in `lit/corcoran_karels_1985` and its Lean file is `CorcoranKarels.lean`; the row is L29 of `MANUSCRIPT.tex`. The Lean file proves the entrant's best reply against committed bets by weighted AM-GM, gives the hardball count the closed form $m \leq r^{r/(r-1)}/(r-1)$, which refutes the claim that only one player survives above $r = 4$, and records misprints in Tables 1 to 4 and in the text.

A file uploaded as "The asymptotic behavior of dynamic rent-seeking games" is Chiarella and Szidarovszky (2002), which studies the local stability of a dynamic rent-seeking game with a fixed number of agents. It is not the paper of Corcoran and Karels and does not cite either of them. Read in full, [**F**]; its record is in `lit/chiarella_szidarovszky_2002` and its Lean file is `ChiarellaSzidarovszky.lean`. It has no entry decision and bears on none of our claims. The record notes that the paper's own second-order condition rules out its Example 2, the example offered for limit cycles, while an unstable complex pair still occurs at other parameters that satisfy the condition.

16 Binary investment before the contest: search of 13 October 2026

C7, C12 and C13 of `MANUSCRIPT.tex` wait on this search. The class searched for is a contest in which agents choose, before the contest, whether to make a costly investment that raises their score, and those who do not invest still compete. The search of 9 October 2026 stopped at abstracts and kept no query log. This one used web search, publisher metadata (Crossref) and abstracts (RePEc). Its queries combined: binary or pre-contest investment with contests, tournaments and rank-order prizes; purchased head starts; statistical discrimination with a binary skill investment and a fixed number of positions; test preparation and capacity-constrained admissions; and strategic classification with unequal costs. The leads fall into five classes.

Same structure: a binary investment, a noisy score, a fixed number of positions, and non-investors who still compete. Fryer and Loury (2013), circulated as Fryer and Loury (2010), is the closest paper found. Agents choose effort $e \in \{0, 1\}$, productivity is a noisy function of effort, a fixed supply of slots goes to the most productive through a market, and identity groups differ in the distribution of investment costs. Fang and Moro (2010) (§6.6.3) state that a group whose cost distribution is higher in the likelihood-ratio order holds a smaller fraction of the slots. That fraction is close to the paying class's share of C10 and C13, and the cost distributions are close to our wealth profile. Their analysis is of policy toward groups (sighted and blind affirmative action). Our analysis is of the count of payers in a finite contest with a rank-order prize. Read in full since, as the working paper (NBER WP 16568) and as the journal's page proofs of 8 June 2013, [**F**]. The published model is the working paper's two-job case: a fixed measure of slots sold at a market price, with a unique laissez-faire investment rate. Its record is in `lit/fryer_loury_2010` and its Lean file is `FryerLoury.lean`; the row is L25 of `MANUSCRIPT.tex`. Under laissez-faire the group with higher costs sits uniformly lower in the hierarchy (p. 11), and a draft passage conjectures that a blind rule lowers investment in both groups (p. 26).

A binary investment and non-investors who still compete, with no fixed number of positions. In Coate and Loury (1993) workers choose whether to invest, a noisy test signals the investment, and employers assign by a threshold, so a worker who did not invest can still get the better job. Read in full, [**F**]; record in `lit/coate_loury_1993`, Lean file `CoateLoury.lean`,

row L28. The number assigned to the better job moves with the standard, which employers set from their beliefs, so the investment rate need not be unique (Proposition 1). In the uniform example, at a fixed employer policy, both investment rates fall as the test becomes less accurate. This is the direction of M28, for the accuracy of a test rather than a weight on talent, and the paper shows it only through the formulas of the example. The assignment rate is split between investors and non-investors and analysed as a function of the standard, not as the investors' share as a function of how many invest. The text layer of footnote 21 misreads $r = 2/3$ as $r = 3$; with 2/3 the footnote's bounds are exactly the hypotheses of Proposition 3. Moro and Norman (2003) make wages competitive. **[C]**, from Fang and Moro (2010).

A fixed number of positions and a continuous investment by everyone. Fryer and Loury (2005), circulated as Fryer and Loury (2003), has pair-wise tournaments in which a blind redistribution rule creates a pool of contestants who exert zero effort, and the pool grows with the aggressiveness of the rule. Read in full since, as the working paper (NBER WP 10104), **[F]**; record in `lit/fryer_loury_2003`, Lean file `FryerLouryTournament.lean`, row L27. The growth of the pool is asserted in the paper; the Lean file derives it from the representation constraint under the hazard-rate order. This extensive margin inside a contest has M28's direction, for a redistribution target rather than a weight on talent, and C12 is narrowed accordingly. In Fu and Lu (2009), neither the pre-contest investment nor effort is monotonic in the number of contestants. Read in full since, as the authors' manuscript of February 2009, **[F]**; record in `lit/fu_lu_2009`, Lean file `FuLu.lean`, row L26. The investment is continuous and every contestant competes, so it is not a precedent for C7. Its non-monotonicity does not hold within the symmetric pure-strategy equilibria it studies: the derivative used for Proposition 1(ii) is misprinted, zero investment for $K \geq 4$ needs $\lambda \geq (K-1)(K+1)^2/(4K^3)$ rather than the printed $2(K-1)^2/K^3$, and the candidate for $K = 3$ always loses to investing nothing. Konrad (2002) studies investment before a contest under incumbency advantages, and Cornell and Welch (1996) rank candidates by exogenous signals of different precision. **[A]** for these, except Fryer and Loury, whose working paper is cached.

A binary action that buys entry or information, with non-takers excluded. In Garg et al. (2026) a student pays to take a test, which reveals information rather than raising the score, and a student who does not take it cannot apply to a school that requires it. The model sections (§§2 and 5.1 of arXiv version 7) were read for this classification. Liu and Lu (2019) study prize allocation in an all-pay auction with costly entry. **[A]**.

Head starts and risk choices, not bought. In Kawamura and Moreno de Barreda (2014) the designer gives a head start, and in Hvide and Kristiansen (2003) contestants choose how much risk to take. **[A]**.

What the search changes. C13 is narrowed by Fryer and Loury to the payers' share as a function of how many pay, in a finite contest with a fixed prize, and its comparative statics; Coate and Loury do not analyse that share. C12 is narrowed by Fryer and Loury (2005) and by Coate and Loury to the sign for a fixed prize awarded by rank and a fixed fee, for every law of talent and of the bought component, and waits on no paper found. C13 waited on Corcoran and Karels (1985), now read: those who do not bet cannot win there, so C13 stands. C7 waits on no paper found: Fu and Lu (2009) is read and is not a precedent. C13's sentence that in every entry model read those who do not enter cannot win stays true of the entry models. It does not cover the statistical-discrimination models, in which those who do not invest can still win.

17 Application search of 14 October 2026

The Research question of `MANUSCRIPT.tex` (§1) rests on this search. Education was dropped as the application because referees read it through human capital. The search looked for settings in which a fixed outlay buys standing in a ranked allocation, those without it still compete, and wealth matters only through affordability. It scoped about 45 settings (log in `TODO.md`). The author chose two areas, bought priority in administrative allocation and the sale of status, and asked for current institutions over historical ones. The two institutions below come from the first area. The literature lists that follow record what the search found, grouped by subject. They are not judged as precedents here, because the conclusions may still change (author, 14 October 2026).

The suite `lit/applications_2026/verify_applications.py` checks every quotation below verbatim, with whitespace removed as the only normalisation. Quotations from the institutions are checked against the cached and pinned documents. Quotations from papers are checked against `lit/applications_2026/abstracts_2026.json`, which holds each paper's abstract as Crossref or OpenAlex returned it on 14 October 2026 and is pinned by sha256. English renderings of the Japanese and Italian are ours.

17.1 The two institutions, from primary sources

Chiba City daycare selection (Chiba City, 2026, 2025). The city states its rule as 「保育の必要性を点数化し、点数の高い方から、入所の決定をします。」 ("the need for childcare is scored, and places are decided from the highest score"). The current criteria rank children first by priority tier, 「優先項目に該当する児童（A）から選考を行い」 ("selection starts with children who meet a priority item (A)"), and within a tier by points: 「選考順位内で入所選考点数が高い方から、希望している保育園等に空きがある場合に、内定します。」 ("within a selection rank, a place is provisionally offered from the highest selection points, where the requested nursery has a vacancy"). One adjustment item covers 「認可外保育施設又はベビーシッター、幼稚園の実施する預かり保育等」 ("an unlicensed nursery or a babysitter, extended care run by a kindergarten, and so on") with the condition 「一時預かり事業の利用(月６４時間以上)」 ("use of temporary care, at least 64 hours a month"), and it scores 「＋４」; the suite checks that the "+4" sits on the same row of the table as the item. The criteria are current: 「令和９年４月利用選考から適用」 ("applied from the selection for April 2027").

Italian public tenders (Repubblica Italiana, 2023, 2006). Article 108(7) of the public contracts code provides that "le stazioni appaltanti prevedono, nei bandi di gara, negli avvisi e negli inviti, il maggior punteggio da attribuire alle imprese per l'adozione di politiche tese al raggiungimento della parità di genere comprovata dal possesso della certificazione della parità di genere" ("contracting authorities provide, in calls for tenders, notices and invitations, for the higher score to be awarded to firms for adopting policies aimed at gender equality, proven by holding the gender-equality certificate"). The certificate exists "al fine di attestare le politiche e le misure concrete adottate dai datori di lavoro per ridurre il divario di genere" ("to attest the policies and concrete measures adopted by employers to reduce the gender gap"), and a decree sets "i parametri minimi per il conseguimento della certificazione della parità di genere" ("the minimum parameters for obtaining the certificate").

Where each institution departs from the model is recorded in `lit/applications_2026/CLAIMS.md` (AP-D1, AP-D2): in Chiba the points are earned by buying a service whose price the provider sets, so the outlay is not one fixed fee; in Italy the certificate needs compliance as well as a payment.

17.2 Literature found, grouped by subject

Each entry quotes its abstract, tagged **[A]** with its source. Two have since been read in full, as marked; the rest have no `lit/` record or Lean yet, and a paper read in full gets both.

Bought priority and allocation of scarce places.

- Sönmez and Switzer (2013): "Cadets at USMA can increase their branch priorities at a fraction of slots by extending their service agreement." [A, OpenAlex]
- Sönmez (2013): "the Army recently introduced incentives programs in which cadets could bid 3 years of additional service obligation to obtain higher priority for their desired branches." [A, OpenAlex]
- Greenberg et al. (2024): "Four objectives— respecting merit, increasing retention, aligning talent, and enhancing trust—guided reforms to the US Army's centralized branching process of cadets to military specialties since 2006." [A, Crossref]
- Kleinrock (1967): "In this paper we consider that relative position in queue is determined according to the size of a customer's bribe (which is paid before the customer sees the queue length)." [A, Crossref]
- Adiri and Yechiali (1974): "The higher the priority of the queue, the higher the toll fee to join it but the shorter the time spent in the system." [A, Crossref]
- Hassin and Haviv (1997): "We provide a detailed analysis of a queueing model in which two priority levels can be purchased." [A, Crossref]; read in full since, from the authors' preprint, **[F, preprint version]**, record `lit/hassin_haviv_1997`, row L31
- Gavirneni and Kulkarni (2016): "Service providers, in the presence of congestion and heterogeneity of customer waiting costs, often introduce a fee-based premier option using which the customers self-segment themselves." [A, Crossref]
- Fernández and Galí (1999): "We compare the performance of markets and tournaments as allocative mechanisms in an economy with borrowing constraints." [A, OpenAlex]; read in full since, as NBER WP 5930, **[F, WP version]**, record `lit/fernandez_gali_1999`, row L30
- Che et al. (2013): "This article studies different methods of assigning a good to budget-constrained agents." [A, OpenAlex]
- Akbarpour et al. (2024): "Many scarce public resources are allocated at below-market-clearing prices, and sometimes for free." [A, OpenAlex]
- Condorelli (2013) and Yamaguchi et al. (2018), the second on childcare rationing in Japan: neither Crossref nor OpenAlex holds an abstract, so they are cited by title only.

Sale of office and status.

- Allen (1998): "This article argues that the system was efficient in that it solved a selection problem and provided appropriate incentives." [A, OpenAlex]
- Crettez et al. (2025): "Venality, i.e., the sale of public positions, was widely used in the judicial sector in France between the 16th and 18th centuries." [A, OpenAlex]
- Guardado (2018): "The paper uses a unique hand-collected dataset of the prices at which the Spanish Crown sold colonial provincial governorships in seventeenth and eighteenth century Peru to examine the impact of colonial officials on long-run development." [A, Crossref]

- Twinam (2015): "Members of mixed races could, however, free themselves from such burdensome restrictions through the purchase of a gracias al sacar —a royal exemption that provided the privileges of Whiteness." [A, OpenAlex]
- Zhang (2013): "Wealth was the key to unlocking this path of advancement and continued to play a role in the competition for success even among those who had already been certified as eligible for office." [A, Crossref]
- Puga and Trefler (2014): "However, starting in 1297, a small group of particularly wealthy merchants blocked political and economic competition: they made parliamentary participation hereditary and erected barriers to participation in the most lucrative aspects of long-distance trade." [A, Crossref]
- Bai and Jia (2016): "This paper studies how the abolition of an elite recruitment system—China's civil exam system that lasted over 1,300 years—affects political stability." [A, OpenAlex]
- Chen et al. (2020): "China's civil examination system (keju), an incredibly long-lived institution, has a persistent impact on human capital outcomes today." [A, Crossref]

Credentials, certification and costly verification.

- Ben-Porath et al. (2014): "There are no monetary transfers, but the principal can check an agent's information at a cost." [A, Crossref]
- Lizzeri (1999): "I show that in a class of environments the optimal choice for a monopoly intermediary is to reveal only whether quality is above some minimal standard." [A, OpenAlex]
- Stahl and Strausz (2017): "While both sellers and buyers have a demand for certification, its role differs fundamentally: sellers use it for signaling, buyers use it for inspection." [A, OpenAlex, WP version]
- Frankel and Kartik (2022): "We formalize why and how a decision maker should commit to underutilizing data." [A, Crossref]
- Stiglitz (1975): "This paper focuses on the labeling of individuals, on the economic costs and benefits of labeling, the institutions that provide it, and the determination of the equilibrium amount of screening under various institutional arrangements." [A, OpenAlex]
- Bidner (2014): "I propose a model in which credentials, such as diplomas, are instrumentally valuable to workers." [A, Crossref]
- Hopkins (2012): "This paper considers a matching model of the labor market where workers, who have private information on their quality, signal to firms that also differ in quality." [A, OpenAlex]
- Bartoš et al. (2016): "In both countries we study, negatively stereotyped minority names reduce employers' effort to inspect resumes." [A, Crossref]
- Spence (1973), Arrow (1973) and Townsend (1979): cited by title only. OpenAlex returns a table of contents for Spence and the opening of the text for Townsend, and no source holds an abstract for Arrow.

The two read in full. Fernández and Galí (1999) and Hassin and Haviv (1997) were read in full on 14 October 2026, because their abstracts sit closest to C1 to C14. Neither changes a claim. In Fernández and Galí wealth matters only through a borrowing constraint, with linear

utility, the wealth distribution is fixed, everyone spends a continuous amount, and who spends nothing gets the lowest school; as the signal weighs ability more, aggregate spending tends to zero (C12). In Hassin and Haviv identical customers buy priority at a fixed toll and those who do not buy are still served; buying follows the crowd, so equilibria can be many, the opposite monotonicity to our count (C3, C6). Their stated bound on the number of pure equilibria is one short when $1/(1-\rho)$ is not an integer, which the record proves from their own recursion. The verdicts are in each record's `NOTES.md`.

What the search records. The Stahl and Strausz entry is the 2017 working paper; its journal version was not confirmed, because the DOI the search first returned resolves to a different paper. No paper in these lists was found to model a selector who chooses the weight on a purchasable credential given a cost of verification. That is a statement about the abstracts read, not a novelty claim, and it enters no C row of `MANUSCRIPT.tex`.

18 Contribution and its precedents, carried from `PROOFS.tex`

This section is carried unchanged in substance from `PROOFS.tex` at its retirement (pass 6, 7 October 2026), with its cross-references turned into the model rows of `MANUSCRIPT.tex`. It is input to the conclusions table and to the novelty verdict, and it is not itself a statement of conclusions.

The claim. For a fixed prize awarded by rank, the effect of resource inequality on *how many people compete* has a determinate sign, and the sign is set by a single observable: whether the marginal entrant stands above or below the pivot of the spread. A *dispersive* spread of the resource distribution — one whose displacement changes sign exactly once, the leading case being the linear mean-preserving spread $w \mapsto \bar{\mu} + \lambda(w - \bar{\mu})$ of M13 — *widens* entry when the marginal competitor is richer than the pivot and *shrinks* it when poorer (M13), strictly so under the boundary condition of M15. What the proof uses is weaker still. The rise branch reads the quantile difference at the marginal entrant alone, the fall branch reads the quantile difference at the first outsider alone, and mean-preservation is nowhere among the hypotheses (M12). Within that class the rule holds globally, distribution-free, and under burden-monotonicity of u alone (M23; strict concavity is sufficient but not necessary, M24).

That class is a genuine restriction, and it belongs with the claim rather than only with the scope discussion at the end of this section. A Rothschild–Stiglitz mean-preserving spread need *not* be dispersive, since its quantile difference may cross zero any number of times. With the marginal entrant above the mean, a change that preserves the total and majorizes the original profile can lower k^* (M13), so the rule stated with the mean in place of the pivot fails for such changes. The sign rule is therefore a theorem about dispersive spreads and about the margin condition, and not about mean-preserving spreads as such, and the rule should not be quoted in the latter form. Which changes the margin condition covers is settled in M14. The condition is silent exactly when the marginal entrant becomes poorer and the first outsider becomes richer. Dispersive changes never reach that band, and inside the band the displacement signs determine nothing, for mean-preserving spreads as for other changes.

The economic content is a reconciliation. The received result — inequality dampens status competition — goes back to Cole et al. (1992), whose §IV.A example has the more compact income distribution producing the higher savings rate, and is developed on the intensive margin by Hopkins and Kornienko (2004) and Hopkins and Kornienko (2010), where less dispersed endowments intensify conspicuous consumption. That finding is here recovered as one branch: mass competitions, in which the marginal entrant sits low in the distribution. Exclusive competitions, in which the marginal entrant is already wealthy, sit on the other branch, where

inequality *feeds* entry. The received result is the below-pivot case of a two-branch rule, not the general case. This is non-tautological because V is fixed by construction (M11): nothing about the prize moves when the distribution spreads.

The margin is endogenous and the rule survives that (M13): the hypothesis is a single comparison at the pre-spread margin, and the wealth ordering carries the rest, so no post-spread margin need be located. Costrell and Loury (2004), the closest structural precedent, work with a marginal quantile fixed by technology. The claim here is not that the present result is the stronger one — they sign a smooth aggregate under a general spread, which is the harder object — but that the extensive-margin formulation replaces that aggregate with a threshold statistic, and an endogenous margin is tractable in the second setting and not the first. The formulation is what is being offered.

Where several entrant sets are equilibria, the assortative one is the cheapest, and under an anonymous gain schedule it therefore maximises the unweighted sum of payoffs (M8) — so using it is a stated selection with an argument behind it, not an assumption. That argument is not a welfare claim, for the reasons set out with the proposition.

Two features make the claim well-posed, and both required proof. The object being signed is an invariant of the game: every pure-strategy equilibrium has the same entrant count (M6), so "how many compete" does not depend on which equilibrium is selected — even though *who* competes does (Appendix D, Information). And the class of spreads for which the rule is proved is not bespoke: a pivot-spread is exactly a dispersive-order shift with a crossing point (M13), the dispersive order being the inequality order this literature already uses (Hopkins and Kornienko, 2009; Shaked, 1982).

Why the extensive margin is the point. The intensive-margin version of the same question exists and is *not* sign-determinate. Schroyen and Treich (2016) sign a mean-preserving spread in wealth inside the same privilege-contest payoff: their Theorem 3 gives the effect as positive iff $2A(1-m^2) > P$, with $A = -u''/u'$ and $P = -u'''/u''$ — local, two players, and conditioned on the contest technology and the *third* derivative of utility. The dependence on u''' is not cosmetic: at $m = 1/2$, CARA with $a = 1$ and log utility at $w = 1$ share $A = 1$ but have $P = 1$ and $P = 2$, and give opposite signs. The extensive-margin rule of M13 needs only that κ is strictly decreasing, which is $u'' < 0$. So the margin is not a modelling convenience; it is what makes inequality's effect on competition determinate at all. An anticipated objection should be met here rather than left to a referee: the two results answer the same economic question — what does a spread of the resource distribution do to competitive expenditure — in different models, and the comparison of hypotheses across margins is legitimate precisely because the margin is the object of choice being defended. It is also not robust terrain for the intensive side: in the assignment literature the direction of the inequality effect reverses when the allocation re-optimises (Costrell and Loury (2004), Proposition 10 against their Proposition 6), so conditioning results of this kind on the margin is already the literature's own practice.

The 1981 precedent. Lazear and Rosen (1981), in the "Income Distributions" subsection of their §III, already have agents self-selecting into a tournament by endowed wealth. This must be acknowledged, with two qualifications that bound what it concedes.

First, their claim is offered as an example and they say so: the subsection opens "While it is not possible to make a general argument based on an example, table 1 suggests…", and rests on one utility function, quadratic costs, normal errors and two endowment levels. They *suggest* the sorting under declining absolute risk aversion; the present model *derives* it, distribution-free.

Second, the channel differs, and decisively. Their sorting runs through DARA, which is a restriction on u''': $A'(w) < 0$ has numerator $(u'')^2 - u'''u'$ and so requires $u''' > 0$. Here the

sorting runs through $\kappa(w) = u(w) - u(w-c)$ being decreasing, which follows from $u'' < 0$ alone. Quadratic utility separates them: it is concave, has $u''' = 0$, and exhibits *increasing* absolute risk aversion — so κ still sorts while DARA fails outright.

Third, and least often noticed, the object of choice differs. Theirs is a choice between two compensation schemes that both pay, so the sorting is tolerance for the binomial spread of a lottery against a concentrated alternative. Ours is whether to pay a fixed entry cost at all, so the sorting is affordability in utility units. That is a difference in the economics, not only in a derivative.

What is not claimed. The mechanism — wealth sorting entry through concavity — is not novel; the two precedents above own it between them, and the sentence claimed here is the two-branch sign rule, not the sorting. The remaining results are apparatus in its service and are presented as such. The uniform cap of M10 is the deterministic, heterogeneous, utility-unit form of an accounting bound already present in Fu et al. (2015) (equation (2) with Definition 2) and Fu and Lu (2010) (equation (7)); it is used here to dispose of the ceiling objection, not claimed as new. The characterisation of M16 to M22 runs on the order-statistic kernel of Ryvkin and Drugov (2020), whose message — that no competitor-number prediction is universal across noise distributions — is theirs in print; what is added is the extensive-margin instance that refutes the referee's conjecture. And the scope of the central claim is stated with it. The rule is proved under the margin condition of M12, which reads the quantile difference at the marginal entrant for the rise branch and at the first outsider for the fall branch. Pivot-spreads are the special case in which one branch applies at every margin. A mean-preserving spread is covered when the displacement is non-negative at the marginal entrant or non-positive at the first outsider, and the displacement may cross zero freely at the other ranks. The condition is a restriction and not a formality, since two mean-preserving spreads with the same displacement signs at every rank can put the margin in the band and move k^* in opposite directions (M14). What is not settled is whether some statistic of the change other than its signs predicts the direction inside the band (TODO item O1).

19 What still needs doing

Read in full as of this version: Hopkins–Kornienko 2004, 2009, 2010; Schroyen–Treich (WP); Ryvkin–Drugov (TE, published); Drugov–Ryvkin (JET, WP); Fu–Jiao–Lu (published); Fu–Lu (WP); Cornes–Hartley (WP); Treich (WP); Levin–Smith (published); Morgan–Orzen–Sefton (published); Moreno–Wooders (published); Che–Gale 1997 (published); Suen (published); Costrell–Loury (published); Cole–Mailath–Postlewaite 1992 (published) and 1995 (WP); Becker–Murphy–Werning (draft); Moldovanu–Sela 2001 (draft); Hoppe–Moldovanu–Sela (published); Lazear–Rosen (published); Schoemaker 1982 (published); Sen 1973 (published); Fullerton–McAfee (published, printed pages 573 to 603); Lewis–Thompson 1981 (published). Every substantive claim in this document about what a paper does now rests on a full read, except where marked [A] or [C]. Where a working paper or draft was read, the published version has not been line-checked. Each paper cited in `PROOFS.tex` now also has a Lean file and a measuring suite under `lit/`, and `lit/COVERAGE.md` lists the coverage.

Done since the previous version. `PROOFS.tex` now states the complete-information assumption, cites Lazear and Rosen (1981) as the precedent for wealth-based self-selection with the u'' against u''' distinction, and cites Cole et al. (1992, 1995) where V is introduced. The comparison of P-MU with Ryvkin and Drugov (2020) has been computed (verdict item 3), and the general-spread question was closed by the margin condition rather than by the Suen and Costrell–Loury technique (§8).

Outstanding, in descending order of consequence for the current claims:

1. The leads of §15 have all been read: Fullerton and McAfee (1999), Mathews and Namoro (2008), Corcoran (1984) and, last, Corcoran and Karels (1985) (13 October 2026). The novelty verdict now fixes the conclusions.
2. Shaked (1982) has been read (9 October 2026; its record is in `lit/shaked_1982` and its Lean file is `ShakedDispersive.lean`); he credits the order to Saunders and Moran (1978) and its name to Lewis and Thompson (1981). Still to do: give the claims made of Hopkins and Kornienko (2004) and Hopkins and Kornienko (2009) a Lean file of their own. Lewis and Thompson (1981), an earlier source for the same order, has been read and covers the definition.
3. Published-version checks: Schroyen–Treich (*GEB*, Theorem 3 and the payoff specifications); Fu–Lu (*EI*, the title differs between MPRA and Fu–Jiao–Lu's citation); Cole–Mailath–Postlewaite 1995 (*QR*, "Concern" or "Concerns"); Becker–Murphy–Werning, Moldovanu–Sela, Cornes–Hartley, Treich, Drugov–Ryvkin (proposition numbering).
4. Grossmann and Dietl (2015) and Chung and Lee (2017), for the entry-condition comparison and the welfare template respectively — still at abstract level.

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